TURRET OS: Multi-Layered Graph and Rule-AST Architecture Rescuing Insider Threat Detection Against Out-of-Distribution Adversaries

Ali Hamza[1,*], **Ghulam Mujtaba**[2]

[1] National University of Sciences and Technology (NUST), Islamabad, Pakistan
Email: ahamza.msse25mcs@student.nust.edu.pk
ORCID: 0009-0006-9790-6643

[2] Department of Computer Science, National University of Computer and Emerging Sciences (FAST-NUCES), Islamabad, Pakistan
Email: i257619@isb.nu.edu.pk
ORCID: 0009-0006-3988-8088

[*] **Corresponding Author:**
Ali Hamza
National University of Sciences and Technology (NUST), Islamabad, Pakistan
Email: ahamza.msse25mcs@student.nust.edu.pk

Target Journal

Computers & Security (Elsevier)

Abstract

Insider-threat detectors fail against OOD adversaries (insiders using stolen credentials during business hours, with low feature novelty, identity-proxy reuse, and removable-media transfers) because single-layer novelty, off-hours, and metadata-driven detectors collapse under this regime. Existing graph-based insider-threat models (PS0, MEWRGNN, DTGI, TGCN-DA, SENTINEL) plateau at ROC-AUC 0.91–0.95 and assume benign traffic clusters tightly; no deployed system achieves full ISO/IEC 27043 forensic-readiness coverage, leaving alerts detached from evidence chains. TURRET OS addresses this gap as a four-layer architecture fusing cross-format metadata harvest, Neo4j knowledge-graph representation, grammar-driven rule-AST inference, and multi-layer GraphSAGE fusion, emitting W3C-PROV, Merkle-chained, Ed25519-signed evidence packs aligned to all eight readiness attributes. On D3 TSEC (60-30-10 chronological split), TURRET OS reports ROC-AUC 0.9997, F1@0.5% FPR 0.8774, and MCC 0.9770; on the held-out D5 `OOD_LAB_EXFILTRATOR`, ROC-AUC 0.9928, F1 0.8936, and MCC 0.9539. The multi-layer stack restores OCC-CP robustness from a 26.25% drop to 4.12% (inside the $\leq 5\%$ bound) while delivering 100% ISO/IEC 27043 coverage at 15.0 s per 1 K activity records.

Keywords

Insider Threat Detection; Provenance Knowledge Graphs; Graph Neural Networks; Digital Forensic Readiness; ISO/IEC 27043; Out-of-Distribution Robustness